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Prof. Taha Havakho

Reddit's Internet Courtroom: Deciding Who's Right or Wrong?

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Prepared By:

Ayda Elzohbi (260899679)

Berly Biju (261208805)

Helia Mahmood Zadeh (261224416)

Rajiha Mehdi (261203537)

Maisy Mofakhami (261178077)

1 Introduction

Reddit’s “Am I the A**hole?” (AITA) subreddit, with over 25 million users, is a hub for crowd-sourced moral judgment. While most models rely on analyzing post content, we take a different route, focusing on user interactions. Using data from the Pushshift Reddit API, we build interaction graphs from thousands of threads and train graph-based classifiers to predict verdicts like YTA or NTA without using any text.

1.1 Objective

This analysis aims to analyze AITA posts using social media analytics to uncover patterns in user interactions, comment structures that influence Reddit’s crowd-sourced moral judgments.

2 Methodology

We transform unstructured Reddit discussions into structured interaction graphs to predict crowd-based moral judgments. After filtering for valid verdicts (NTA, YTA, ESH, NAH), we build user–post interaction graphs and extract features using PageRank, centrality metrics, and community detection. These graph-based features are used to train models like Random Forest and XGBoost. By relying solely on interaction patterns—not text—we show that social dynamics alone can effectively forecast moral judgments online.

2.1 Data Collection & Processing

The data collection process began by retrieving readily-available data from the [Pushshift Reddit API](#), selecting both the comments (9.2 GB) and submissions (414 MB) files from [r/AmItheAsshole](#). A balanced set of submissions was sampled across verdict categories, and relevant comments were sampled. Chunked reading was used, as well as reducing sample size and task splitting to avoid long runtimes.

2.2 Graph Construction

For each post, we constructed a user–post interaction graph where nodes represent users and edges represent comment interactions. The goal was to model the structural properties of discussions, capturing how individuals respond to one another in these moral debates. We computed several graph metrics, including: Tree depth and branching factor, Clustering coefficients, PageRank and betweenness centrality, and Temporal dynamics of comment chains. These metrics helped quantify influence, discussion density, and user positioning without relying on semantic content.

2.3 Feature Selection and Classification

Using graph-derived features, we trained multiple classification models to predict AITA verdicts. For the test and train split, we divided the whole graph into training and test sets. Random Forest was used for interpretability and robustness, XGBoost for performance tuning, and deep learning models like Graph Neural Networks (GNNs) and Graph Convolutional Networks (GCNs) to capture spatial node influence within the interaction graphs. We applied community detection algorithms to identify echo chambers and modular structures, and used centrality metrics to pinpoint influential users driving community verdicts. Throughout our workflow, we experimented with different modeling strategies: starting from binary classification (e.g., YTA vs. NTA), scaling to multi-class classification across all four verdicts (YTA, NTA, ESH, NAH), and finally comparing performance across classical ML models and graph-based neural

architectures. This iterative approach allowed us to evaluate trade-offs in accuracy, interpretability, and scalability across diverse methodologies.

3 Results and Findings

3.1 Classification Performance & Network Patterns

Our best-performing model was XGBoost with optimized hyperparameters, achieving a binary classification accuracy of ~71% and the highest ROC-AUC score, outperforming baseline and multiclass models. The binary setup (YTA vs. NTA) delivered the most robust predictive performance, suggesting that the model could effectively separate polarized judgments based solely on interaction dynamics.

We also experimented with multi-class classification, text-based features, and multi-modal models (combining graph and text features), but interaction-based features alone consistently held strong. Techniques like SMOTE were applied to handle class imbalance, especially in multiclass setups, though they didn't significantly outperform the simpler binary structure. Behaviorally, we found:

- High-centrality users (via PageRank, eigenvector) were closely aligned with final verdicts, highlighting influencer effects.
- Branched, dense conversation trees often led to ESH outcomes, reflecting diffuse or contentious debates.
- Early commenters had higher PageRank scores, suggesting a timing advantage in shaping judgment.

These results reinforce our core hypothesis: social interaction patterns alone can predict moral judgments, without any reliance on textual sentiment or semantics.

3.2 Community Detection & Centrality Effects

We used Louvain community detection to segment discussions into clusters, revealing strong echo chamber effects, where users within tight-knit subgroups frequently reinforced the same verdict direction. Networks with high modularity tended to exhibit greater alignment and signs of groupthink, suggesting that moral consensus often forms within isolated clusters. Centrality analysis further highlighted the role of network position in shaping outcomes. Users with high betweenness centrality acted as bridges between these clusters, sometimes introducing competing viewpoints or disrupting consensus formation. Meanwhile, eigenvector and degree centrality were strongly correlated with upvote counts, indicating that user visibility played a significant role in their influence over the final verdict. Altogether, these findings suggest that crowd-based moral judgments on Reddit are not only socially constructed but also disproportionately shaped by a small number of strategically positioned users and the structure of the interaction network itself.

4 Conclusion

This project demonstrates that structured social behaviours offer predictive signals for crowd-sourced moral judgment. By shifting the analytical lens from what users say to how they interact, we unlock a new approach to modelling digital discourse. Our graph-based system not only delivers strong predictive performance but also contributes to broader discussions on the mechanics of online influence, judgment formation, and community-driven decision making. Future directions may include integrating sentiment as a secondary layer, analyzing cross-subreddit generalizability, and applying the model to real-time moderation or debate forecasting.